

Nike-shorts-tiktok-insights

2025 Capstone Project Documentation & README

1. Project Overview

Welcome to the official repository for the **Nike-shorts-tiktok-insights** capstone project. As part of the Nike Global Digital Strategy Team, this project explores influencer marketing and creator-led short-form content across TikTok and YouTube Shorts to extract actionable marketing insights, evaluate campaign performance, and build predictive models.

2. Repository Structure & File Manifest

- `nike_capstone.sql` : Contains all 12 analytical SQL queries covering platform comparisons, regional hotspots, creator impact, hashtag ROI, emoji effects, upload timing, traffic breakdown, device brand analysis, and data quality checks.
- `import_data.py` : Automated Python ingestion script using Pandas and SQLAlchemy to load raw CSV data into a local MySQL database instance.
- `S NIKE DASHBOARD.pbix` : Power BI multi-page interactive dashboard covering Executive Overview, Content Insights, Creator Insights, and Market/Device Analysis.
- **Datasets & Outputs:**
 - `youtube_shorts_tiktok_trends_2025.csv` : Raw primary dataset.
 - `youtube_shorts_tiktok_trends_2025.csv_ML.csv` : Feature-engineered dataset utilized for machine learning regression pipelines.
 - `country_platform_summary_2025.csv` , `monthly_trends_2025.csv` , `top_creators_impact_2025.csv` , `top_hashtags_2025.csv` : Aggregated summary exports.
- `DATA_DICTIONARY.csv` : Column definitions and schema documentation.

3. Phase 1: SQL Analytics (`nike_capstone.sql`)

The MySQL script handles schema initialization and implements 12 core business intelligence queries:

- **Platform Comparison:** Evaluates total views and average engagement rates across YouTube Shorts vs. TikTok.
- **Regional Hotspots:** Extracts the top 5 countries by view count per platform using Common Table Expressions (CTEs) and window ranking functions (`RANK( ) OVER`).
- **Category Performance:** Analyzes average completion rates and watch times across sports, lifestyle, and fashion categories.
- **Creator Impact:** Identifies the top 10 author handles based on average views alongside their creator tier.
- **Hashtag ROI:** Computes top hashtags by total views alongside their median engagement rates using median estimation subqueries.
- **Emoji Effect:** Compares engagement metrics between videos featuring emojis in titles versus those without.
- **Upload Timing:** Analyzes optimal upload hours and days of the week based on view counts and completion rates.
- **Traffic Source Performance:** Breaks down engagement velocity and trend duration by trend type.

- **Device Brand Analysis:** Evaluates completion performance grouped by device type and brand.
- **Traffic Sources Breakdown:** Aggregates video volume, average views, and median completion rates per traffic source.
- **Seasonal Insights:** Compares engagement patterns across event seasons (Olympics, Christmas, Summer sales).
- **Data Quality Check:** Audits mathematical integrity by verifying whether `engagement_total = views + likes + shares + saves`.

4. Phase 2: Automated Ingestion Pipeline (`import_data.py`)

To stream the raw data into your MySQL database workbench, run the Python automation script:

```
import pandas as pd
from sqlalchemy import create_engine

# 1. Read CSV using pandas
df = pd.read_csv("youtube_shorts_tiktok_trends_2025.csv")

# 2. Database credentials
username = "root"
password = "your_password"
host = "localhost"
port = "3306"
database = "nike_capstone"

# 3. Create Connection & Push to SQL
engine = create_engine(
    f"mysql+mysqlconnector://{username}:{password}@{host}:{port}/{database}?charset=utf8mb4"
)
df.to_sql(
    name="youtube_shorts_tiktok_trends_2025.csv",
    con=engine,
    if_exists="replace",
    index=False,
)
print("Import complete! Refresh MySQL Workbench to view tables.")
```

5. Phase 3: Power BI Dashboards (`s_NIKE_DASHBOARD.pbix`)

The interactive business intelligence report is organized across multiple comprehensive pages:

- **Executive Overview:** High-level KPI cards (Total Views, Total Videos, Average Engagement Rate, Average Completion Rate) with cross-filtering support for Platform, Region, Country, and Category.
- **Content Insights:** Granular analysis of video categories, hashtag performance, emoji effects, duration, and watch-time distributions.
- **Creator Insights:** Rankings of top creators by tier, upload timing heatmaps (hour/day), and detailed engagement breakdowns (like rate, share rate, save rate).

- **Market & Device Analysis:** Geographic views by country/region, device type/brand completion metrics, and traffic source comparisons.

6. Phase 4: Machine Learning & Predictive Modeling

Using `youtube_shorts_tiktok_trends_2025.csv_ML.csv`, the machine learning pipeline focuses on:

- **Feature Engineering:** Processing text keywords, sound types, category mappings, and numerical durations.
- **Model Training:** Implementing regression models to forecast engagement velocity and audience retention.
- **Evaluation:** Validating predictive performance via cross-validation and error metrics (R^2 and RMSE).

7. How to Push to GitHub

Execute the following commands in your project terminal:

```
git init
git add .
git commit -m "Complete Nike-shorts-tiktok-insights capstone project files and documentation"
git branch -M main
git remote add origin https://github.com/YOUR_USERNAME/Nike-shorts-tiktok-insights.git
git push -u origin main
```